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MIL-M-28787/500(SH)
29 DECEMBER 1989

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MIL-STD-1397 PARALLEL CHANNEL CONTROL MODULE, KEY CODE AWB

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic MIL-STD-1397 parallel channel control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.466 failures per million operating hours for class I or 4.376 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be AWB.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
A2-A11	Isolation	II	X	10	INF	10	INF	10	INF	MOhm
B1-B8	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Power supply stress	IV	S	----------Module shall pass all remaining tests----------						
D1-D5	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D6	Ibd	V	S	-0.12	1.12	-0.12	1.12	-0.12	1.12	mA
D7	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D8	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D9	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D10	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D11-D14	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D15-D17	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D18	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D19	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D20	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D21	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D22,D23	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D24	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
E1	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E2-E6	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E7	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E8	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E9-E18	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E19	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E20-E36	Ios	VI	S	-220.0	-30.0	-220.0	-30.0	-220.0	-30.0	mA
E37	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E38-E53	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 + 5°C 60 ± 5°C		Class II 4/ -55 + 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F1	Ith	VII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
F2-F6	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F7	Ith	VII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
F8	Ith	VII	X	-0.4	0.03	-0.4	0.03	-0.4	0.03	mA
F9	Ith	VII	X	-0.2	0.01	-0.2	0.01	-0.2	0.01	mA
F10	Ith	VII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
F11	Ith	VII	X	-0.2	0.00	-0.2	0.00	-0.2	0.00	mA
F12	Ith	VII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
F13-F15	Ith	VII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
F16-F18	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F19	Ith	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F20	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F21	Ith	VII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
F22	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F23,F24	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F25	Ith	VII	X	-4.0	0.1	-4.0	0.1	-4.0	0.1	mA
F26-F41	Ith	VII	X	-0.22	0.04	-0.22	0.04	-0.22	0.04	mA
F42-F57	Ith	VII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
G1	Ith	VIII	X	-0.4	0.04	-0.4	0.04	-0.4	0.04	mA
G2-G6	Ith	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G7	Ith	VIII	X	-7.43	-2.82	-7.43	-2.82	-7.43	-2.82	mA
G8	Ith	VIII	X	-0.4	0.03	-0.4	0.03	-0.4	0.03	mA
G9	Ith	VIII	X	-0.2	0.01	-0.2	0.01	-0.2	0.01	mA
G10	Ith	VIII	X	-0.4	0.02	-0.4	0.02	-0.4	0.02	mA
G11	Ith	VIII	X	-0.2	0.00	-0.2	0.00	-0.2	0.00	mA
G12	Ith	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G13-G15	Ith	VIII	X	-5.63	-2.85	-5.63	-2.85	-5.63	-2.85	mA
G16-G18	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G19	Ith	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G20	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G21	Ith	VIII	X	-0.8	0.04	-0.8	0.04	-0.8	0.04	mA
G22	Ith	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G23,G24	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G25	Iil	VIII	X	-4 0	0 1	-4.0	0.1	-4.0	0 1	mA
G26-G41	Iil	VIII	X	-0 22	0 04	-0.22	0.04	-0.22	0 04	mA
G42-G57	Iil	VIII	X	-0.25	0.07	-0.25	0.07	-0.25	0.07	mA
H1-H53	Voh	IX	X	2 4	4.5	2.4	4.5	2.4	4 5	V
J1-J61	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0 5	V
K	Iozh	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	mA
L1-L9	Vic	XII	S	-1 5	0 0	-1.5	0.0	-1.5	0 0	V
L10-L19	Vic	XII	S	-1 2	0.0	-1.2	0.0	-1.2	0 0	V
M1	PSFC	XIII	S	38.7	68.1	38.7	68.1	38.7	68.1	uF
M2	PSFC	XIII	S	9.9	18.5	9.9	18.5	9.9	18.5	uF
N	Icc	XIV	X	164	393	164	393	164	393	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R2-R6	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R7-R10	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R11-R13	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R14	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R15	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R16	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R17-R19	C1/D	XVIII	Q	2	75	2	75	2	75	pF
R20-R22	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R23	C1/D	XVIII	Q	4	75	4	75	4	75	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R24	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R25,R26	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R27,R28	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R29-R36	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R37-R39	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R40-R44	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R45	C1/D	XVIII	Q	4	125	4	125	4	125	pF
R46-R52	C1/D	XVIII	Q	4	100	4	100	4	100	pF
R53	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R54	C1/D	XVIII	Q	6	75	6	75	6	75	pF
R55	C1/D	XVIII	Q	4	75	4	75	4	75	pF
R56	C1/D	XVIII	Q	8	100	8	100	8	100	pF
R57	C1/D	XVIII	Q	6	100	6	100	6	100	pF
T	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/500-1 for class I and M28787/500-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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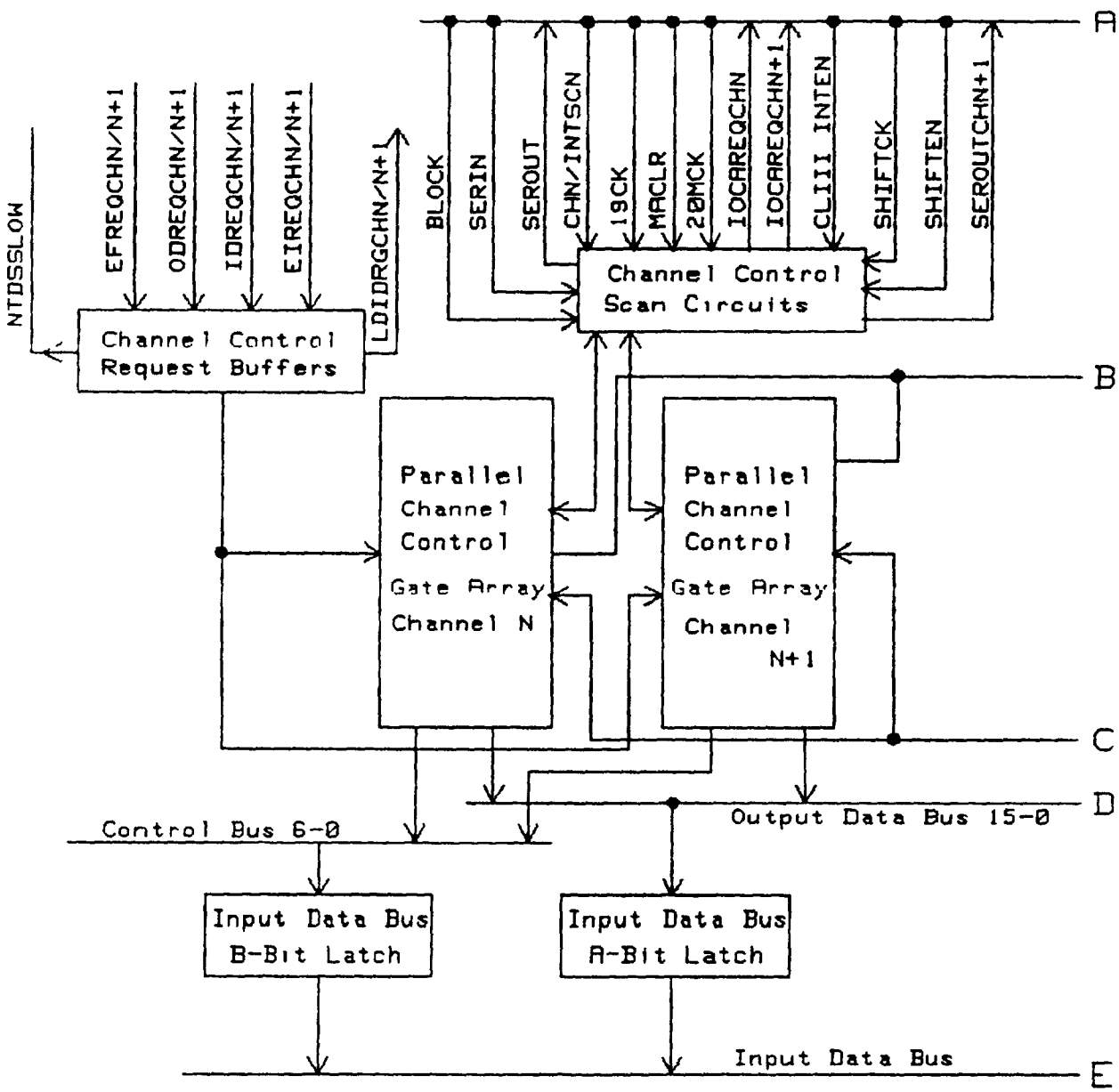


FIGURE 1. AWB functional block diagram.

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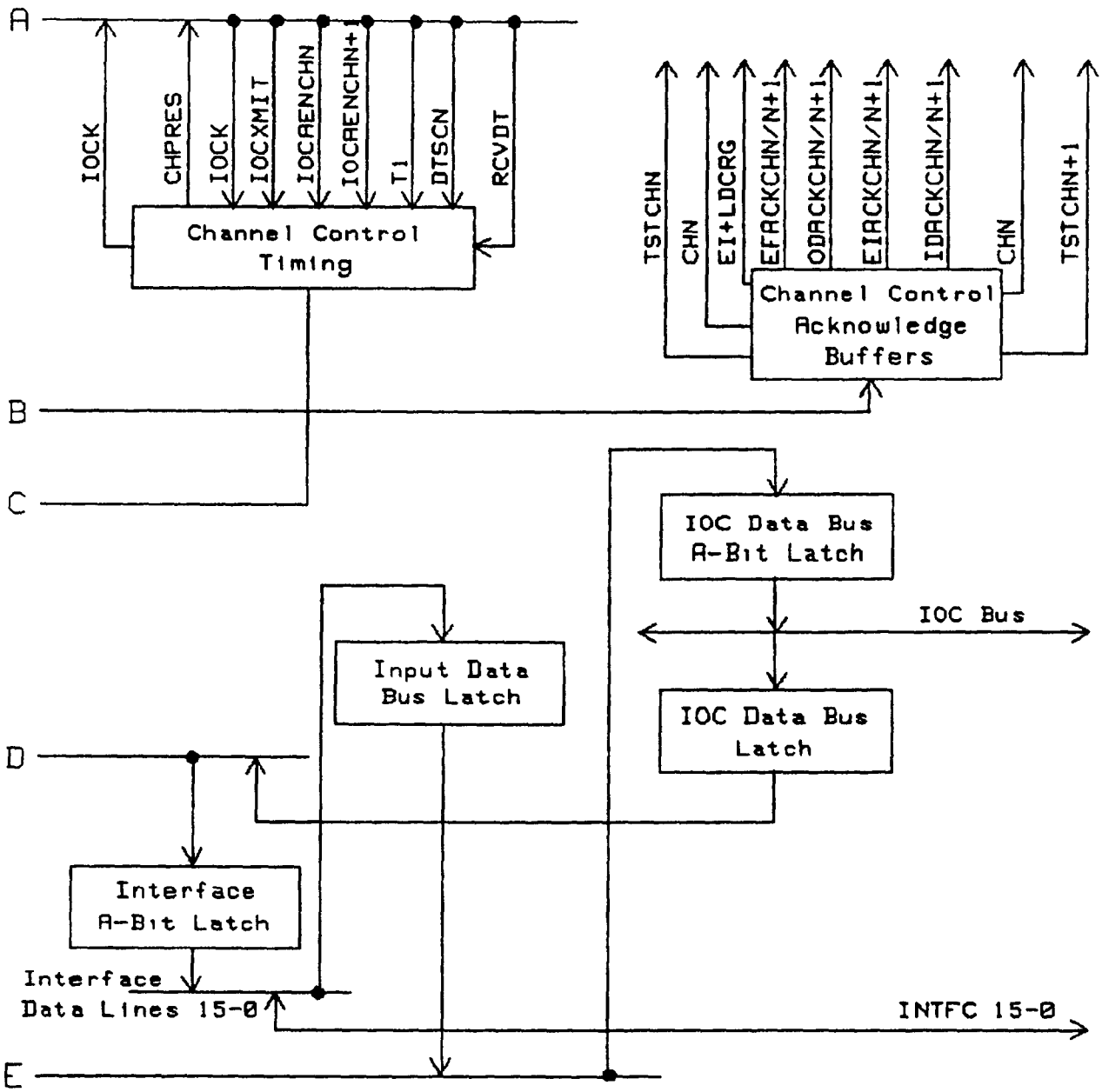


FIGURE 1. AWB functional block diagram - Continued.

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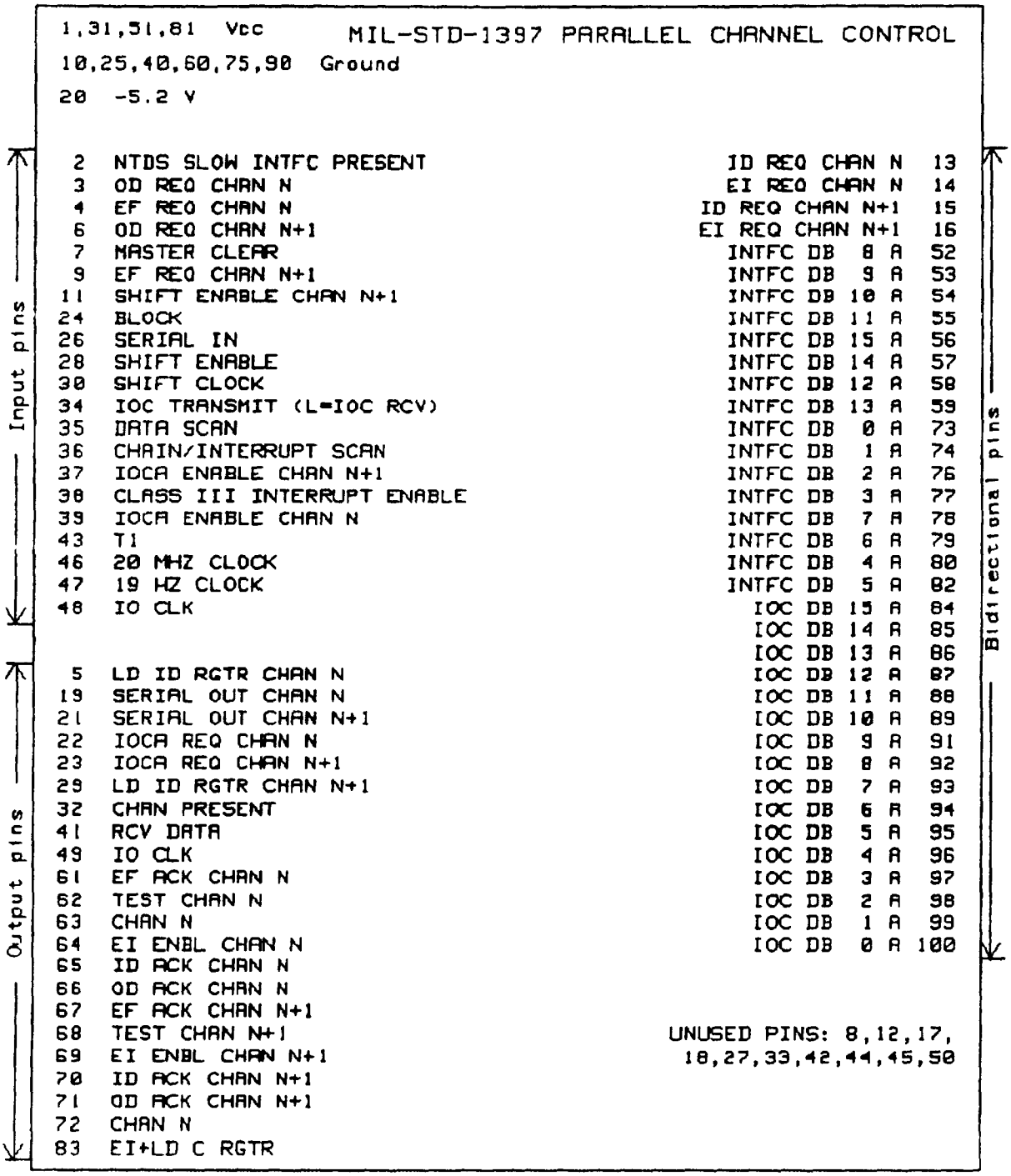


FIGURE 2. AWB pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

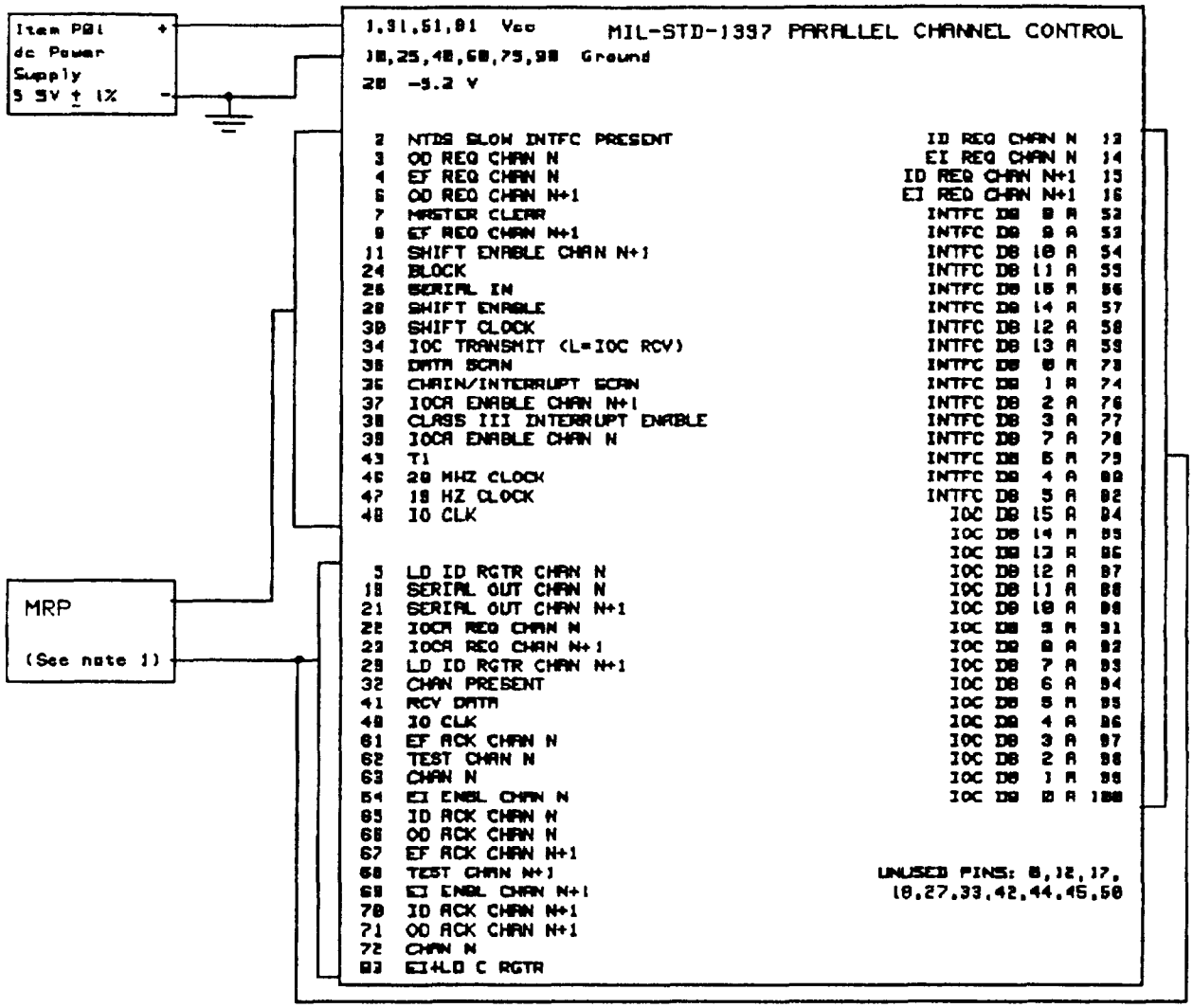
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. AWB operating temperature and component temperature test circuit.

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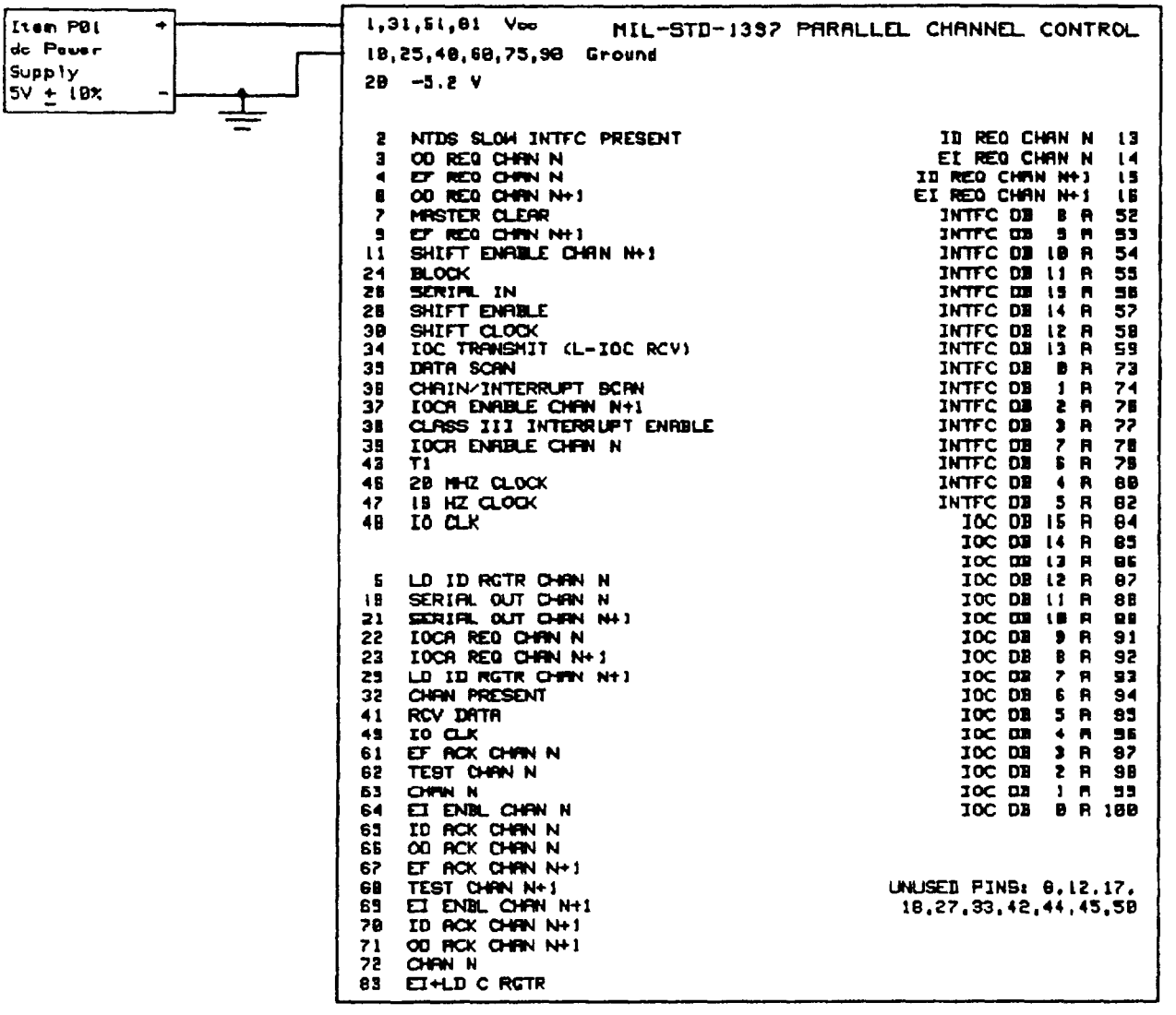


FIGURE 4. AWB vibration test circuit.

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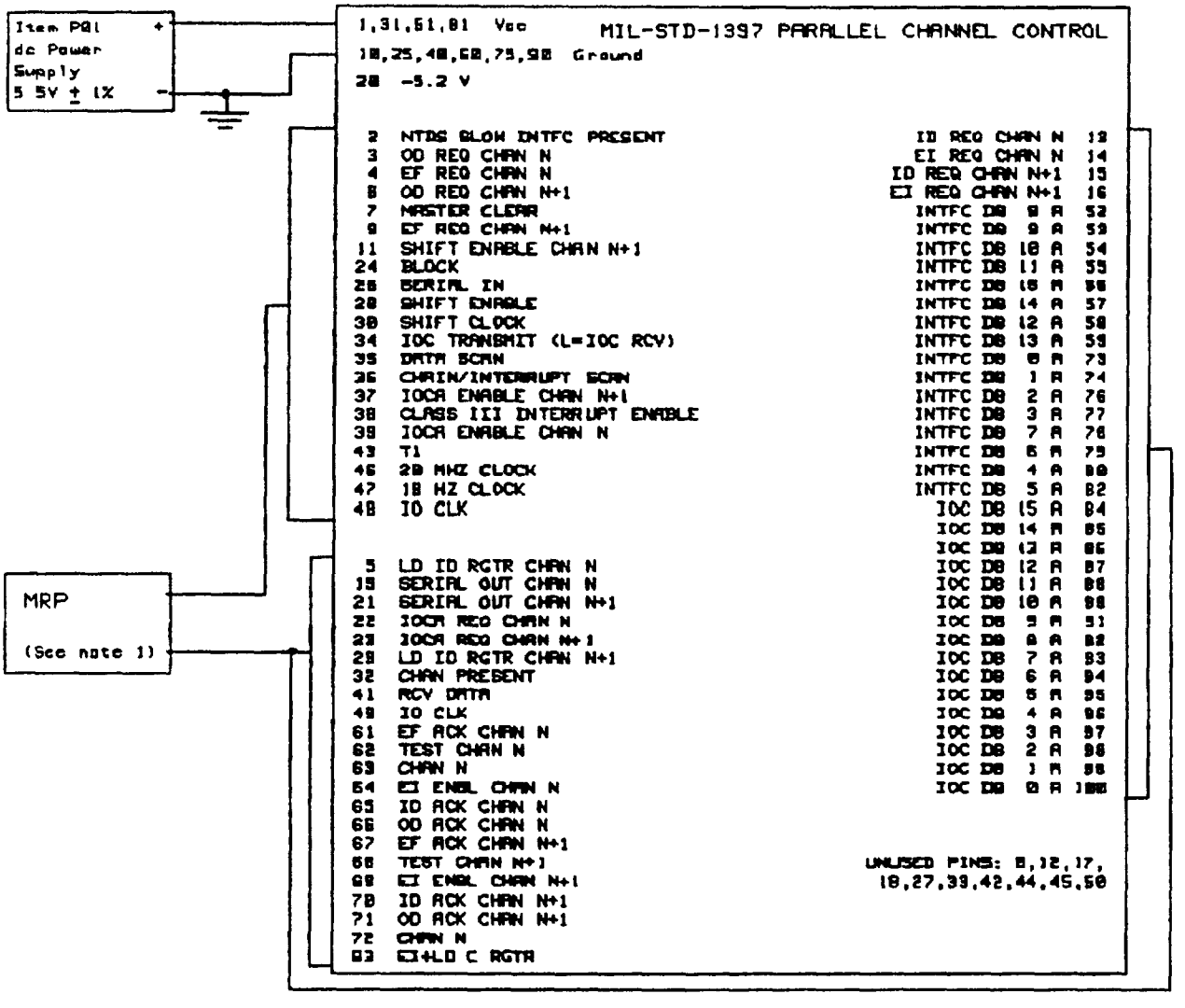
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. AWB life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 2.16 W (max); 1.45 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
NTDSSLOW	NA	0.4	2.4	-0.4	0.04	NA	100
ODREQCHN	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EFREQCHN	-1.5	0.4	2.4	-0.2	0.02	0.1	75
ODREQCHN+1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
MACLR	-1.5	0.4	2.4	-0.2	0.02	0.1	75
EFREQCHN+1	-1.5	0.4	2.4	-0.2	0.02	0.1	75
SHIFTENCHN+1	NA	0.4	2.4	-7.43	-2.82	1.12	75
IDREQCHN	-1.5	0.4	2.4	-0.4	0.03	0.2	75
EIREQCHN	-1.5	0.4	2.4	-0.2	0.01	0.1	75
IDREQCHN+1	-1.5	0.4	2.4	-0.4	0.02	0.2	100
EIREQCHN+1	-1.5	0.4	2.4	-0.2	0.00	0.1	100
BLOCK	NA	0.4	2.4	-6.03	-2.85	0.22	75
SERIN	NA	0.4	2.4	-5.63	-2.85	0.22	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
SHIFTEN	NA	0.4	2.4	-5.63	-2.85	0.22	75
SHIFTCK	NA	0.4	2.4	-5.63	-2.85	0.22	75
IOCXMIT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
DTSCN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CHN/INTSCN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOCAENCHN+1	-1.2	0.4	2.4	-0.8	0.04	0.2	100
CLIII INTEN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
IOCAENCHN	-1.2	0.4	2.4	-0.8	0.04	0.2	100
T1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
20MCK	-1.2	0.4	2.4	-2.0	0.05	1.0	75
19CK	-1.2	0.4	2.4	-2.0	0.05	1.0	75
IOCK	-1.2	0.4	2.4	-4.0	0.1	2.0	100
INTFC 8	NA	0.4	2.4	-0.22	0.04	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
INTFC 9	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 10	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 11	NA	0.4	2.4	-0.22	0.04	NA	75
INTFC 15	NA	0.4	2.4	-0.22	0.04	NA	75
INTFC 14	NA	0.4	2.4	-0.22	0.04	NA	75
INTFC 12	NA	0.4	2.4	-0.22	0.04	NA	75
INTFC 13	NA	0.4	2.4	-0.22	0.04	NA	75
INTFC 0	NA	0.4	2.4	-0.22	0.04	NA	125
INTFC 1	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 2	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 3	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 7	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 6	NA	0.4	2.4	-0.22	0.04	NA	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
INTFC 4	NA	0.4	2.4	-0.22	0.04	NA	100
INTFC 5	NA	0.4	2.4	-0.22	0.04	NA	100
IOC 15	NA	0.4	2.4	-0.25	0.07	NA	100
IOC 14	NA	0.4	2.4	-0.25	0.07	NA	100
IOC 13	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 12	NA	0.4	2.4	-0.25	0.07	NA	100
IOC 11	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 10	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 9	NA	0.4	2.4	-0.25	0.07	NA	100
IOC 8	NA	0.4	2.4	-0.25	0.07	NA	100
IOC 7	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 6	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 5	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IOC 4	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 3	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 2	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 1	NA	0.4	2.4	-0.25	0.07	NA	75
IOC 0	NA	0.4	2.4	-0.25	0.07	NA	75

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
LDIDRGCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IDREQCHN	NA	0.5	NA	1.6	NA	NA	NA	75
EIREQCHN	NA	0.5	NA	1.6	NA	NA	NA	75
IDREQCHN+1	NA	0.5	NA	1.6	NA	NA	NA	100
EIREQCHN+1	NA	0.5	NA	1.6	NA	NA	NA	100
SEROUTCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
SEROUTCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IOCAREQCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IOCAREQCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
LDIDRGCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CHPRES	NA	0.5	NA	1.6	0.5	NA	NA	NA
RCVDT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
IOCK	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
INTFC 8	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 9	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 10	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 11	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
INTFC 15	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
INTFC 14	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
INTFC 12	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
INTFC 13	2.4	0.5	-0.16	1.6	NA	NA	-30.0	75
EFACKCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
TSTCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
EIENCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IDACKCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ODACKCHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
EFACKCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
TSTCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
EIENCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IDACKCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
ODACKCHN+1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CHN	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
INTFC 0	2.4	0.5	-0.16	1.6	NA	NA	-30.0	125
INTFC 1	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 2	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 3	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 7	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 6	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
INTFC 4	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
INTFC 5	2.4	0.5	-0.16	1.6	NA	NA	-30.0	100
EI+LDCRG	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
IOC 15	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 14	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 13	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 11	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 10	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 9	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 8	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
IOC 7	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 6	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
IOC 5	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75
IOC 0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	75

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	IOCK to IOC 0	-	-	-	86	86	86	nS	12
1/ 1	tpd	IOCK to IOC 1	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 2	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 3	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 4	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 5	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 6	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 7	-	-	-	86	86	86	nS	12
2	tpd	IOCK to IOC 0	-	-	-	70	70	70	nS	12
2/ 2	tpd	IOCK to IOC 1	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 2	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 3	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 4	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 5	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 6	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 7	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 8	-	-	-	70	70	70	nS	12

1/ These signals contain control information.
2/ These signals contain data.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	IOCK to IOC 9	-	-	-	70	70	70	nS	12
2/ 2	tpd	IOCK to IOC 10	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 11	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 12	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 13	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 14	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 15	-	-	-	70	70	70	nS	12
3	tpd	IOCK to INTFC 0	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 1	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 2	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 3	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 4	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 5	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 6	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 7	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 8	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 9	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 10	-	-	-	73	73	73	nS	12

2/ These signals contain data.

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	IOCK to INTFC 11	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 12	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 13	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 14	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 15	-	-	-	73	73	73	nS	12
4	tpd	IOCK to IOCAREQCHN	-	-	-	83	83	83	nS	12
4	tpd	IOCK to IOCAREQCHN+1	-	-	-	83	83	83	nS	12
5	tpd	IOCK to IDACKCHN	-	-	-	110	110	110	nS	12
5	tpd	IOCK to IDACKCHN+1	-	-	-	110	110	110	nS	12
6	tsu	T1 to IOCK	25	25	25	-	-	-	nS	12
7	th	IOCK to T1	4	4	4	-	-	-	nS	12
8	tpd	20MCK to ODACKCHN	-	-	-	103	103	103	nS	12
8	tpd	20MCK to ODACKCHN+1	-	-	-	103	103	103	nS	12
8	tpd	20MCK to EFACKCHN	-	-	-	103	103	103	nS	12
8	tpd	20MCK to EFACKCHN+1	-	-	-	103	103	103	nS	12

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6.4 Module function. The parallel channel control module contains the Complementary Metal Oxide Semiconductor (CMOS) 1397 parallel channel control microcircuit for channel N and channel N+1, channel control scan circuits, request buffers and channel control timing to capture data from the various data and control busses.

6.4.1 Channel control scan circuits. The channel control scan circuits consist primarily of inverters used to buffer the scan and clock signals destined for the parallel channel control gate array. The 20 MHz clock from the Input/Output Controller (IOC) data bus upper module (EWC) is applied to a divide-by-two flip-flop to provide the CLOCK A input to the gate array.

6.4.2 Channel control request buffers. The channel control request buffers are composed of inverter circuits for External Function Request (EFR), Output Data Request (ODR), External Interrupt Request (EIR) and Input Data Request (IDR) signals for both channel N and channel N+1. Test signals for channel N and channel N+1 are also applied to the request buffer circuits for module tests.

6.4.3 Channel control timing circuits. Channel control timing receives timing and enable signals from the IOC channel control module (AWH) and the IOC common bus control modules (BWF) as inputs. Enable signals are buffered, gated and delayed to provide enables for channel N and N+1 in addition to data IN and OUT signals. Timing phase T1 is delayed and gated to provide additional T2, T3, and T4 timing phases. The IOC clock is buffered and used to clock the timing/enable flip-flop and for various clocking requirements throughout the module.

6.4.4 Input data bus latches. The input data bus latches are the input data bus A-bit latch and the input data bus B-bit latch. The A-bit latch receives data from the parallel channel gate array chip outputs and latches it on the input data bus, bits 15-0. Enables are provided by the Input/Output (I/O) clock and control is exercised by the Not Data Scan - T4 signal from the channel control gate array control bus bits 6-0. Clocking is done by the I/O clock with enables at T3 time. Outputs from the B-bit latch are placed on input data bus bits 7-0.

6.4.5 Interface latch. The constantly enabled interface data bus A-bit latch receives the output data bus A-bits and places them on the input data bus at time T2.

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6.4.6 IOC data bus latches. Data to be placed on the IOC data bus is transferred via the IOC data bus A-bit latch. Input is from the input data bus with enables from channel control timing circuits. Data applied to the gate array data bus is transferred via the IOC data bus latch. Inputs to the latch are from the IOC data bus with enables from channel control timing circuits.

6.4.7 Input data bus latch. The input data bus latch receives interface bus bits 15-0 and transfers them to the input data bus. Enable is from the channel control timing circuits (Data Scan - T4).

6.4.8 Parallel channel control gate array. The 1397 parallel channel control gate array is a 64-pin microcircuit that performs the control operations for a parallel two-channel group (channel N and N+1). There are two parallel channel control gate arrays per module. The gate array performs:

a) command transfers,

b) chain/interrupt transfers,

c) single-word and dual-word output data transfers,

d) single-word and dual-word input data transfers,

e) single-word and dual-word External Interrupt (EI) transfers, and

f) Externally Specified Addressing (ESA) input and output data transfers.

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6.4.9 Description of I/O pins.

Signal name	Pin number	I/O	Description
NTDSSLOW	2	I	NTDS SLOW INTFC PRESENT
ODREQCHN	3	I	OD REQ CHAN N
EFREQCHN	4	I	EF REQ CHAN N
LDIDRGCHN	5	O	LD ID RGTR CHAN N
ODREQCHN+1	6	I	OD REQ CHAN N+1
MACLR	7	I	MASTER CLEAR
EFREQCHN+1	9	I	EF REQ CHAN N+1
SHIFTENCHN+1	11	I	SHIFT ENABLE CHAN N+1
IDREQCHN	13	I/O	ID REQ CHAN N
EIREQCHN	14	I/O	EI REQ CHAN N
IDREQCHN+1	15	I/O	ID REQ CHAN N+1
EIREQCHN+1	16	I/O	EI REQ CHAN N+1
SEROUTCHN	19	O	SERIAL OUT CHAN N
SEROUTCHN+1	21	O	SERIAL OUT CHAN N+1
IOCAREQCHN	22	O	IOCA REQ CHAN N
IOCAREQCHN+1	23	O	IOCA REQ CHAN N+1
BLOCK	24	I	BLOCK
SERIN	26	I	SERIAL IN
SHIFTEN	28	I	SHIFT ENABLE
LDIDRGCHN+1	29	O	LD ID RGTR CHAN N+1

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
SHIFTCK	30	I	SHIFT CLOCK
CHPRES	32	O	CHAN PRESENT
IOCXMIT	34	I	IOC TRANSMIT (L = IOC RCV)
DTSCN	35	I	DATA SCAN
CHN/INTSCN	36	I	CHAIN/INTERRUPT SCAN
IOCAENCHN+1	37	I	IOCA ENABLE CHAN N+1
CLIII INTEN	38	I	CLASS III INTERRUPT ENABLE
IOCAENCHN	39	I	IOCA ENABLE CHAN N
RCVDT	41	O	RCV DATA
T1	43	I	T1
20MCK	46	I	20 MHZ CLOCK
19CK	47	I	19 HZ CLOCK
IOCK	48	I	IO CLK
IOCK	49	O	IO CLK
INTFC 8	52	I/O	INTFC DB 8 A
INTFC 9	53	I/O	INTFC DB 9 A
INTFC 10	54	I/O	INTFC DB 10 A
INTFC 11	55	I/O	INTFC DB 11 A
INTFC 15	56	I/O	INTFC DB 15 A
INTFC 14	57	I/O	INTFC DB 14 A

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
INTFC 12	58	I/O	INTFC DB 12 A
INTFC 13	59	I/O	INTFC DB 13 A
EFACKCHN	61	O	EF ACK CHAN N
TSTCHN	62	O	TEST CHAN N
CHN	63	O	CHAN N
EIENCHN	64	O	EI ENBL CHAN N
IDACKCHN	65	O	ID ACK CHAN N
ODACKCHN	66	O	OD ACK CHAN N
EFACKCHN+1	67	O	EF ACK CHAN N+1
TSTCHN+1	68	O	TEST CHAN N+1
EIENCHN+1	69	O	EI ENBL CHAN N+1
IDACKCHN+1	70	O	ID ACK CHAN N+1
ODACKCHN+1	71	O	OD ACK CHAN N+1
CHN	72	O	CHAN N
INTFC 0	73	I/O	INTFC DB 0 A
INTFC 1	74	I/O	INTFC DB 1 A
INTFC 2	76	I/O	INTFC DB 2 A
INTFC 3	77	I/O	INTFC DB 3 A
INTFC 7	78	I/O	INTFC DB 7 A
INTFC 6	79	I/O	INTFC DB 6 A

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
INTFC 4	80	I/O	INTFC DB 4 A
INTFC 5	82	I/O	INTFC DB 5 A
EI+LDCRG	83	O	EI+LD C RGTR
IOC 15	84	I/O	IOC DB 15 A
IOC 14	85	I/O	IOC DB 14 A
IOC 13	86	I/O	IOC DB 13 A
IOC 12	87	I/O	IOC DB 12 A
IOC 11	88	I/O	IOC DB 11 A
IOC 10	89	I/O	IOC DB 10 A
IOC 9	91	I/O	IOC DB 9 A
IOC 8	92	I/O	IOC DB 8 A
IOC 7	93	I/O	IOC DB 7 A
IOC 6	94	I/O	IOC DB 6 A
IOC 5	95	I/O	IOC DB 5 A
IOC 4	96	I/O	IOC DB 4 A
IOC 3	97	I/O	IOC DB 3 A
IOC 2	98	I/O	IOC DB 2 A

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6.4.9 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IOC 1	99	I/O	IOC DB 1 A
IOC 0	100	I/O	IOC DB 0 A
-5.2V	20	-	-5.2 volts is not used on this module. The pin, however, is bussed with -5.2 V pins on other modules and acts as added filter capacitance.

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N733)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	C,8, 12,17, 18,27, 33,42, 44,45, 50	Frame	10	INF	MOhm	1-8
A2	Isolation, used to unused pins	C	8,12,17, 18,27, 33,42, 44,45,50	10	INF	MOhm	1-8
A3	Isolation, unused to unused pins	8	12,17, 18,27, 33,42, 44,45,50				
A4	Isolation, unused to unused pins	12	17,18, 27,33, 42,44, 45,50				
A5	Isolation, unused to unused pins	17	18,27, 33,42, 44,45,50				
A6	Isolation, unused to unused pins	18	27,33, 42,44, 45,50				
A7	Isolation, unused to unused pins	27	33,42, 44,45,50				
A8	Isolation, unused to unused pins	33	42,44 45,50				
A9	Isolation, unused to unused pins	42	44,45,50	10	INF	MOhm	1-8

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A10	Isolation, unused to unused pins	44	45,50	10	INF	MOhm	1-8
A11	Isolation, unused to unused pins	45	50	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A11
Isolation measurement	NA	± 500 KOhm	A1-A11

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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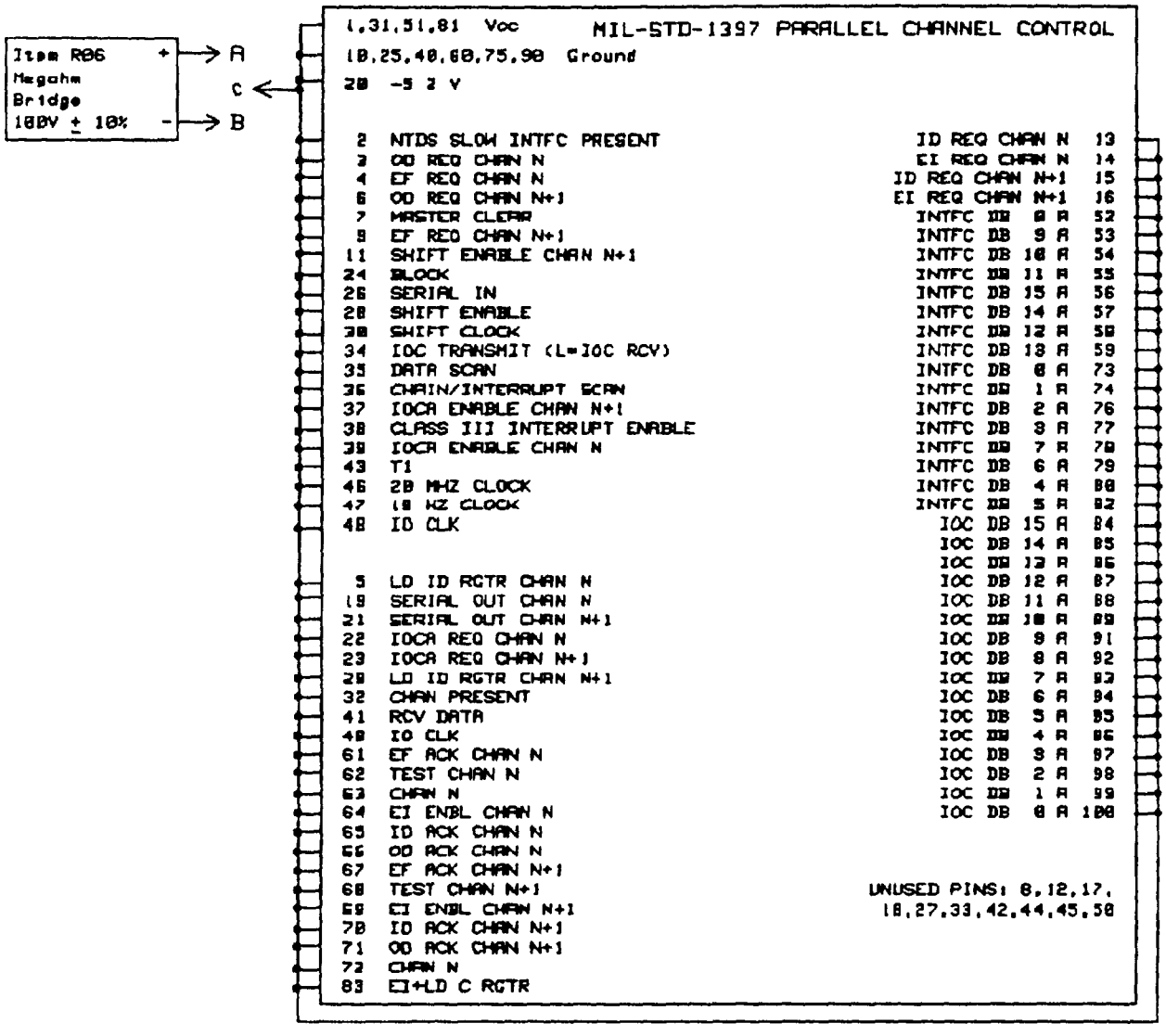


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, GND	10	60				
B6	Continuity, Vcc	1	31				
B7	Continuity, Vcc	51	81				
B8	Continuity, Vcc	1	51	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B8

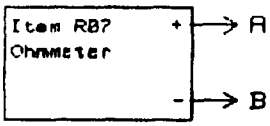
2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,61,81 Vcc
10,25,40,60,75,90 Ground
20 -5.2 V

MIL-STD-1397 PARALLEL CHANNEL CONTROL

2	NTDS SLOW INTFC PRESENT	ID REQ CHAN N	13
3	OD REQ CHAN N	EI REQ CHAN N	14
4	EF REQ CHAN N	ID REQ CHAN N+1	15
6	OD REQ CHAN N+1	EI REQ CHAN N+1	16
7	MASTER CLEAR	INTFC DB 8 A	52
9	EF REQ CHAN N+1	INTFC DB 9 A	53
11	SHIFT ENABLE CHAN N+1	INTFC DB 10 A	54
24	BLOCK	INTFC DB 11 A	55
26	SERIAL IN	INTFC DB 15 A	56
28	SHIFT ENABLE	INTFC DB 14 A	57
30	SHIFT CLOCK	INTFC DB 12 A	58
34	IOC TRANSMIT (L-IOC RCV)	INTFC DB 13 A	59
35	DATA SCAN	INTFC DB 0 A	73
36	CHAIN/INTERRUPT SCAN	INTFC DB 1 A	74
37	IOCA ENABLE CHAN N+1	INTFC DB 2 A	76
38	CLASS III INTERRUPT ENABLE	INTFC DB 3 A	77
39	IOCA ENABLE CHAN N	INTFC DB 7 A	78
43	T1	INTFC DB 6 A	79
46	20 MHZ CLOCK	INTFC DB 4 A	80
47	19 HZ CLOCK	INTFC DB 5 A	82
48	IO CLK	IOC DB 15 A	84
		IOC DB 14 A	85
		IOC DB 13 A	86
5	LD ID RGTR CHAN N	IOC DB 12 A	87
19	SERIAL OUT CHAN N	IOC DB 11 A	88
21	SERIAL OUT CHAN N+1	IOC DB 10 A	89
22	IOCA REQ CHAN N	IOC DB 9 A	91
23	IOCA REQ CHAN N+1	IOC DB 8 A	92
29	LD ID RGTR CHAN N+1	IOC DB 7 A	93
32	CHAN PRESENT	IOC DB 6 A	94
41	RCV DATA	IOC DB 5 A	95
49	IO CLK	IOC DB 4 A	96
61	EF ACK CHAN N	IOC DB 3 A	97
62	TEST CHAN N	IOC DB 2 A	98
63	CHAN N	IOC DB 1 A	99
64	EI ENBL CHAN N	IOC DB 0 A	100
65	ID ACK CHAN N		
66	OD ACK CHAN N		
67	EF ACK CHAN N+1		
68	TEST CHAN N+1	UNUSED PINS: 8,12,17,	
69	EI ENBL CHAN N+1	18,27,33,42,44,45,50	
70	ID ACK CHAN N+1		
71	OD ACK CHAN N+1		
72	CHAN N		
83	EI+LD C RGTR		

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	0.707 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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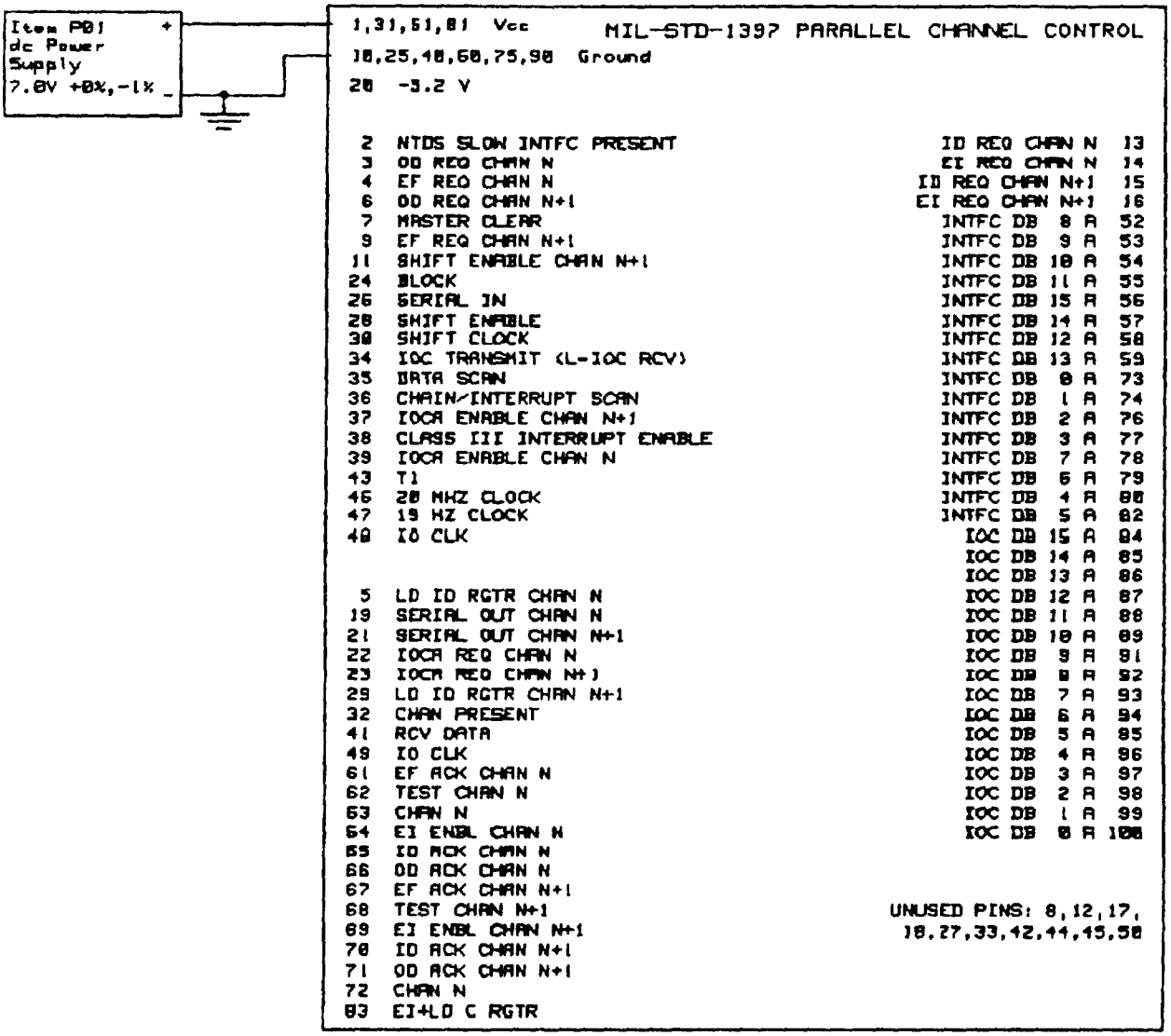


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	ODREQCHN	3	5.5	0.0	0.1	mA	1-3
D2	EFREQCHN	4					
D3	ODREQCHN+1	6					
D4	MACLR	7					
D5	EFREQCHN+1	9	5.5	0.0	0.1	mA	1-3
D6	SHIFTENCHN+1	11	5.5	-0.12	1.12	mA	1-3
D7	IDREQCHN	13	5.5	0.0	0.2	mA	1-3
D8	EIREQCHN	14	5.5	0.0	0.1	mA	1-3
D9	IDREQCHN+1	15	5.5	0.0	0.2	mA	1-3
D10	EIREQCHN+1	16	5.5	0.0	0.1	mA	1-3
D11	BLOCK	24	5.5	-0.12	0.22	mA	1-3
D12	SERIN	26					
D13	SHIFTEN	28					
D14	SHIFTCK	30	5.5	-0.12	0.22	mA	1-3
D15	IOCXMIT	34	5.5	0.0	0.1	mA	1-3
D16	DTSCN	35					
D17	CHN/INTSCN	36	5.5	0.0	0.1	mA	1-3
D18	IOCAENCHN+1	37	5.5	0.0	0.2	mA	1-3
D19	CLIII INTEN	38	5.5	0.0	0.1	mA	1-3
D20	IOCAENCHN	39	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	T1	43	5.5	0.0	0.1	mA	1-3
D22	20MCK	46	5.5	0.0	1.0	mA	1-3
D23	19CK	47	5.5	0.0	1.0	mA	1-3
D24	IOCK	48	5.5	0.0	2.0	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D24
Power supply current limit	0.589 A	NA	D1-D24
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D24
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	D1-D24 D1-D24

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 5 uA	D1-D5
	NA	± 62 uA	D6
	NA	± 10 uA	D7
	NA	± 5 uA	D8
	NA	± 10 uA	D9
	NA	± 5 uA	D10
	NA	± 17 uA	D11-D14
	NA	± 5 uA	D15-D17
	NA	± 10 uA	D18
	NA	± 5 uA	D19
	NA	± 10 uA	D20
	NA	± 5 uA	D21
	NA	± 50 uA	D22,D23
	NA	± 100 uA	D24

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	LDIDRGCHN	5	0.0	-260.0	-30.0	mA	1-4
E2	SEROUTCHN	19	0.0	-220.0	-30.0	mA	1-4
E3	SEROUTCHN+1	21					
E4	IOCAREQCHN	22					
E5	IOCAREQCHN+1	23					
E6	LDIDRGCHN+1	29	0.0	-220.0	-30.0	mA	1-4
E7	RCVDT	41	0.0	-150.0	-60.0	mA	1-4
E8	IOCK	49	0.0	-100.0	-40.0	mA	1-4
E9	INTFC 8	52	0.0	-220.0	-30.0	mA	1-4
E10	INTFC 9	53					
E11	INTFC 10	54					
E12	INTFC 11	55					
E13	INTFC 15	56					
E14	INTFC 14	57					
E15	INTFC 12	58					
E16	INTFC 13	59					
E17	EFACKCHN	61					
E18	TSTCHN	62	0.0	-220.0	-30.0	mA	1-4
E19	CHN	63	0.0	-260.0	-30.0	mA	1-4
E20	EIENCHN	64	0.0	-220.0	-30.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	IDACKCHN	65	0.0	-220.0	-30.0	mA	1-4
E22	ODACKCHN	66					
E23	EFACKCHN+1	67					
E24	TSTCHN+1	68					
E25	EIENCHN+1	69					
E26	IDACKCHN+1	70					
E27	ODACKCHN+1	71					
E28	CHN	72					
E29	INTFC 0	73					
E30	INTFC 1	74					
E31	INTFC 2	76					
E32	INTFC 3	77					
E33	INTFC 7	78					
E34	INTFC 6	79					
E35	INTFC 4	80					
E36	INTFC 5	82	0.0	-220.0	-30.0	mA	1-4
E37	EI+LDCRG	83	0.0	-260.0	-30.0	mA	1-4
E38	IOC 15	84	0.0	-150.0	-60.0	mA	1-4
E39	IOC 14	85					
E40	IOC 13	86	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	IOC 12	87	0.0	-150.0	-60.0	mA	1-4
E42	IOC 11	88					
E43	IOC 10	89					
E44	IOC 9	91					
E45	IOC 8	92					
E46	IOC 7	93					
E47	IOC 6	94					
E48	IOC 5	95					
E49	IOC 4	96					
E50	IOC 3	97					
E51	IOC 2	98					
E52	IOC 1	99					
E53	IOC 0	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E53
Power supply current limit	0.589 A	NA	E1-E53
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E53
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E53
low, (VL)	0.0 V	± 100 mV	E1-E53
Output current measurement	NA	± 11.5 mA	E1
	NA	± 9.5 mA	E2-E6
	NA	± 4.5 mA	E7
	NA	± 3.0 mA	E8
	NA	± 9.5 mA	E9-E18
	NA	± 11.5 mA	E19
	NA	± 9.5 mA	E20-E36
	NA	± 11.5 mA	E37
	NA	± 4.5 mA	E38-E53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	NTDSSLOW	2	2.4	-0.4	0.04	mA	1-3
F2	ODREQCHN	3	2.4	-0.2	0.02	mA	1-3
F3	EFREQCHN	4					
F4	ODREQCHN+1	6					
F5	MACLR	7					
F6	EFREQCHN+1	9	2.4	-0.2	0.02	mA	1-3
F7	SHIFTENCHN+1	11	2.4	-7.43	-2.82	mA	1-3
F8	IDREQCHN	13	2.4	-0.4	0.03	mA	1-3
F9	EIREQCHN	14	2.4	-0.2	0.01	mA	1-3
F10	IDREQCHN+1	15	2.4	-0.4	0.02	mA	1-3
F11	EIREQCHN+1	16	2.4	-0.2	0.00	mA	1-3
F12	BLOCK	24	2.4	-6.03	-2.85	mA	1-3
F13	SERIN	26	2.4	-5.63	-2.85	mA	1-3
F14	SHIFTEN	28					
F15	SHIFTCK	30	2.4	-5.63	-2.85	mA	1-3
F16	IOCXMIT	34	2.4	-0.6	0.02	mA	1-3
F17	DTSCN	35					
F18	CHN/INTSCN	36	2.4	-0.6	0.02	mA	1-3
F19	IOCAENCHN+1	37	2.4	-0.8	0.04	mA	1-3
F20	CLIII INTEN	38	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	IOCAENCHN	39	2.4	-0.8	0.04	mA	1-3
F22	T1	43	2.4	-0.6	0.02	mA	1-3
F23	20MCK	46	2.4	-2.0	0.05	mA	1-3
F24	19CK	47	2.4	-2.0	0.05	mA	1-3
F25	IOCK	48	2.4	-4.0	0.1	mA	1-3
F26	INTFC 8	52	2.4	-0.22	0.04	mA	1-3
F27	INTFC 9	53					
F28	INTFC 10	54					
F29	INTFC 11	55					
F30	INTFC 15	56					
F31	INTFC 14	57					
F32	INTFC 12	58					
F33	INTFC 13	59					
F34	INTFC 0	73					
F35	INTFC 1	74					
F36	INTFC 2	76					
F37	INTFC 3	77					
F38	INTFC 7	78					
F39	INTFC 6	79					
F40	INTFC 4	80	2.4	-0.22	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	INTFC 5	82	2.4	-0.22	0.04	mA	1-3
F42	IOC 15	84	2.4	-0.25	0.07	mA	1-3
F43	IOC 14	85					
F44	IOC 13	86					
F45	IOC 12	87					
F46	IOC 11	88					
F47	IOC 10	89					
F48	IOC 9	91					
F49	IOC 8	92					
F50	IOC 7	93					
F51	IOC 6	94					
F52	IOC 5	95					
F53	IOC 4	96					
F54	IOC 3	97					
F55	IOC 2	98					
F56	IOC 1	99					
F57	IOC 0	100	2.4	-0.25	0.07	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:20Z
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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F57
Power supply current limit	0.589 A	NA	F1-F57
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F57
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F57 F1-F57
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 22.0 uA ± 11.0 uA ± 230.0 uA ± 21.5 uA ± 10.5 uA ± 21.0 uA ± 10.0 uA ± 159.0 uA ± 139.0 uA ± 31.0 uA ± 42.0 uA ± 31.0 uA ± 42.0 uA ± 31.0 uA ± 102.0 uA ± 205.0 uA ± 13.0 uA ± 16.0 uA	F1 F2-F6 F7 F8 F9 F10 F11 F12 F13-F15 F16-F18 F19 F20 F21 F22 F23,F24 F25 F26-F41 F42-F57

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:20Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	NTDSSLOW	2	0.4	-0.4	0.04	mA	1-3
G2	ODREQCHN	3	0.4	-0.2	0.02	mA	1-3
G3	EFREQCHN	4					
G4	ODREQCHN+1	6					
G5	MACLR	7					
G6	EFREQCHN+1	9	0.4	-0.2	0.02	mA	1-3
G7	SHIFTENCHN+1	11	0.4	-7.43	-2.82	mA	1-3
G8	IDREQCHN	13	0.4	-0.4	0.03	mA	1-3
G9	EIREQCHN	14	0.4	-0.2	0.01	mA	1-3
G10	IDREQCHN+1	15	0.4	-0.4	0.02	mA	1-3
G11	EIREQCHN+1	16	0.4	-0.2	0.00	mA	1-3
G12	BLOCK	24	0.4	-6.03	-2.85	mA	1-3
G13	SERIN	26	0.4	-5.63	-2.85	mA	1-3
G14	SHIFTEN	28					
G15	SHIFTCK	30	0.4	-5.63	-2.85	mA	1-3
G16	IOCXMIT	34	0.4	-0.6	0.02	mA	1-3
G17	DTSCN	35					
G18	CHN/INTSCN	36	0.4	-0.6	0.02	mA	1-3
G19	IOCAENCHN+1	37	0.4	-0.8	0.04	mA	1-3
G20	CLIII INTEN	38	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	IOCAENCHN	39	0.4	-0.8	0.04	mA	1-3
G22	T1	43	0.4	-0.6	0.02	mA	1-3
G23	20MCK	46	0.4	-2.0	0.05	mA	1-3
G24	19CK	47	0.4	-2.0	0.05	mA	1-3
G25	IOCK	48	0.4	-4.0	0.1	mA	1-3
G26	INTFC 8	52	0.4	-0.22	0.04	mA	1-3
G27	INTFC 9	53					
G28	INTFC 10	54					
G29	INTFC 11	55					
G30	INTFC 15	56					
G31	INTFC 14	57					
G32	INTFC 12	58					
G33	INTFC 13	59					
G34	INTFC 0	73					
G35	INTFC 1	74					
G36	INTFC 2	76					
G37	INTFC 3	77					
G38	INTFC 7	78					
G39	INTFC 6	79					
G40	INTFC 4	80	0.4	-0.22	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	INTFC 5	82	0.4	-0.22	0.04	mA	1-3
G42	IOC 15	84	0.4	-0.25	0.07	mA	1-3
G43	IOC 14	85					
G44	IOC 13	86					
G45	IOC 12	87					
G46	IOC 11	88					
G47	IOC 10	89					
G48	IOC 9	91					
G49	IOC 8	92					
G50	IOC 7	93					
G51	IOC 6	94					
G52	IOC 5	95					
G53	IOC 4	96					
G54	IOC 3	97					
G55	IOC 2	98					
G56	IOC 1	99					
G57	IOC 0	100	0.4	-0.25	0.07	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:20Z
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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G57
Power supply current limit	0.589 A	NA	G1-G57
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G57
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G57
low, (VL)	0.0 V	± 100 mV	G1-G57
Input current measurement	NA	± 22.0 uA	G1
	NA	± 11.0 uA	G2-G6
	NA	± 230.0 uA	G7
	NA	± 21.5 uA	G8
	NA	± 10.5 uA	G9
	NA	± 21.0 uA	G10
	NA	± 10.0 uA	G11
	NA	± 159.0 uA	G12
	NA	± 139.0 uA	G13-G15
	NA	± 31.0 uA	G16-G18
	NA	± 42.0 uA	G19
	NA	± 31.0 uA	G20
	NA	± 42.0 uA	G21
	NA	± 31.0 uA	G22
	NA	± 102.0 uA	G23,G24
	NA	± 205.0 uA	G25
	NA	± 13.0 uA	G26-G41
	NA	± 16.0 uA	G42-G57

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	LDIDRGCHN	5	-0.16	2.4	4.5	V	1-4
H2	SEROUTCHN	19					
H3	SEROUTCHN+1	21					
H4	IOCAREQCHN	22					
H5	IOCAREQCHN+1	23					
H6	LDIDRGCHN+1	29	-0.16	2.4	4.5	V	1-4
H7	RCVDT	41	-0.4	2.4	4.5	V	1-4
H8	IOCK	49	-0.4	2.4	4.5	V	1-4
H9	INTFC 8	52	-0.16	2.4	4.5	V	1-4
H10	INTFC 9	53					
H11	INTFC 10	54					
H12	INTFC 11	55					
H13	INTFC 15	56					
H14	INTFC 14	57					
H15	INTFC 12	58					
H16	INTFC 13	59					
H17	EFACKCHN	61					
H18	TSTCHN	62					
H19	CHN	63					
H20	EIENCHN	64	-0.16	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	IDACKCHN	65	-0.16	2.4	4.5	V	1-4
H22	ODACKCHN	66					
H23	EFACKCHN+1	67					
H24	TSTCHN+1	68					
H25	EIENCHN+1	69					
H26	IDACKCHN+1	70					
H27	ODACKCHN+1	71					
H28	CHN	72					
H29	INTFC 0	73					
H30	INTFC 1	74					
H31	INTFC 2	76					
H32	INTFC 3	77					
H33	INTFC 7	78					
H34	INTFC 6	79					
H35	INTFC 4	80					
H36	INTFC 5	82					
H37	EI+LDCRG	83	-0.16	2.4	4.5	V	1-4
H38	IOC 15	84	-0.4	2.4	4.5	V	1-4
H39	IOC 14	85					
H40	IOC 13	86	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H41	IOC 12	87	-0.4	2.4	4.5	V	1-4
H42	IOC 11	88					
H43	IOC 10	89					
H44	IOC 9	91					
H45	IOC 8	92					
H46	IOC 7	93					
H47	IOC 6	94					
H48	IOC 5	95					
H49	IOC 4	96					
H50	IOC 3	97					
H51	IOC 2	98					
H52	IOC 1	99					
H53	IOC 0	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H53
Power supply current limit	0.589 A	NA	H1-H53
Forcing current, (Ioh)	-0.16 mA -0.4 mA -0.16 mA -0.4 mA	± 16 uA ± 40 uA ± 16 uA ± 40 uA	H1-H6 H7,H8 H9-H37 H38-H53
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H53 H1-H53
Output voltage measurement	NA	± 105 mV	H1-H53

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

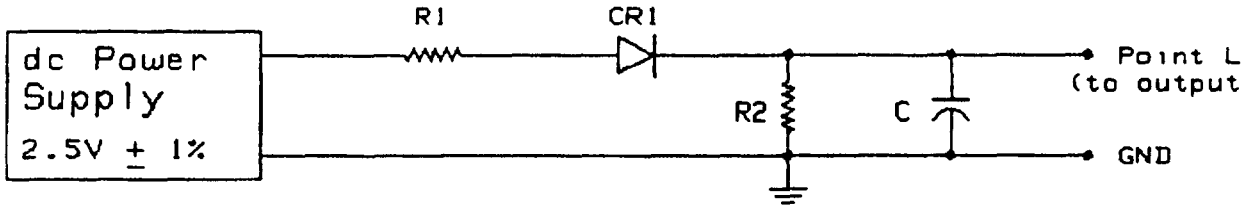
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
5,19,21,22,23,29, 52,53,54,55,56, 57,58,59,61,62, 63,64,65,66,67, 68,69,70,71,72, 73,74,76,77,78, 79,80,82,83	16000	1.6	0.5
13,14,15,16,32	NA	1.6	0.5
41,49,84,85,86, 87,88,89,91,92, 93,94,95,96,97, 98,99,100	6200	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J1	LDIDRGCHN	5	1.6	0.0	0.5	V	1-4
J2	IDREQCHN	13					
J3	EIREQCHN	14					
J4	IDREQCHN+1	15					
J5	IDREQCHN+1	15					
J6	EIREQCHN+1	16					
J7	EIREQCHN+1	16					
J8	SEROUTCHN	19					
J9	SEROUTCHN+1	21					
J10	IOCAREQCHN	22					
J11	IOCAREQCHN+1	23					
J12	LDIDRGCHN+1	29					
J13	CHPRES	32					
J14	CHPRES	32	1.6	0.0	0.5	V	1-4
J15	RCVDT	41	6.0	0.0	0.5	V	1-4
J16	IOCK	49	6.0	0.0	0.5	V	1-4
J17	INTFC 8	52	1.6	0.0	0.5	V	1-4
J18	INTFC 9	53					
J19	INTFC 10	54					
J20	INTFC 11	55	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J21	INTFC 15	56	1.6	0.0	0.5	V	1-4
J22	INTFC 14	57					
J23	INTFC 12	58					
J24	INTFC 13	59					
J25	EFACKCHN	61					
J26	TSTCHN	62					
J27	CHN	63					
J28	EIENCHN	64					
J29	IDACKCHN	65					
J30	ODACKCHN	66					
J31	EFACKCHN+1	67					
J32	TSTCHN+1	68					
J33	EIENCHN+1	69					
J34	IDACKCHN+1	70					
J35	ODACKCHN+1	71					
J36	CHN	72					
J37	INTFC 0	73					
J38	INTFC 1	74					
J39	INTFC 2	76					
J40	INTFC 3	77	1.6	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	INTFC 7	78	1.6	0.0	0.5	V	1-4
J42	INTFC 6	79					
J43	INTFC 4	80					
J44	INTFC 5	82					
J45	EI+LDCRG	83	1.6	0.0	0.5	V	1-4
J46	IOC 15	84	6.0	0.0	0.5	V	1-4
J47	IOC 14	85					
J48	IOC 13	86					
J49	IOC 12	87					
J50	IOC 11	88					
J51	IOC 10	89					
J52	IOC 9	91					
J53	IOC 8	92					
J54	IOC 7	93					
J55	IOC 6	94					
J56	IOC 5	95					
J57	IOC 4	96					
J58	IOC 3	97					
J59	IOC 2	98					
J60	IOC 1	99					
J61	IOC 0	100	6.0	0.0	0.5	V	1-4

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:20Z
Check the source to verify that this is the current version before use.

NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J61
Power supply current limit	0.589 A	NA	J1-J61
Forcing current, (Iol)	1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 160 uA ± 600 uA ± 160 uA ± 600 uA	J1-J14 J15,J16 J17-J45 J46-J61
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J61 J1-J61
Output voltage measurement	NA	± 25 mV	J1-J61

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:20Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
K	CHPRES	32	5.5	0.0	0.5	mA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K
Power supply current limit	0.589 A	NA	K
Forcing voltage, (Voh)	5.5 V	± 50 mV	K
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	K K
Output current measurement	NA	± 25 uA	K

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	ODREQCHN	3	-18.0	-1.5	0.0	V	1-3
L2	EFREQCHN	4					
L3	ODREQCHN+1	6					
L4	MACLR	7					
L5	EFREQCHN+1	9					
L6	IDREQCHN	13					
L7	EIREQCHN	14					
L8	IDREQCHN+1	15					
L9	EIREQCHN+1	16	-18.0	-1.5	0.0	V	1-3
L10	IOCXMIT	34	-18.0	-1.2	0.0	V	1-3
L11	DTSCN	35					
L12	CHN/INTSCN	36					
L13	IOCAENCHN+1	37					
L14	CLIII INTEN	38					
L15	IOCAENCHN	39					
L16	T1	43					
L17	20MCK	46					
L18	19CK	47					
L19	IOCK	48	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L19
Power supply current limit	NA	NA	L1-L19
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L19
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L19 L1-L19
Input voltage measurement	 NA NA	 ± 75 mV ± 60 mV	 L1-L9 L10-L19

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	36.7	68.1	uF	1-4
M2	PSFC	20	9.9	18.5	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 1.57 uF ± 0.43 uF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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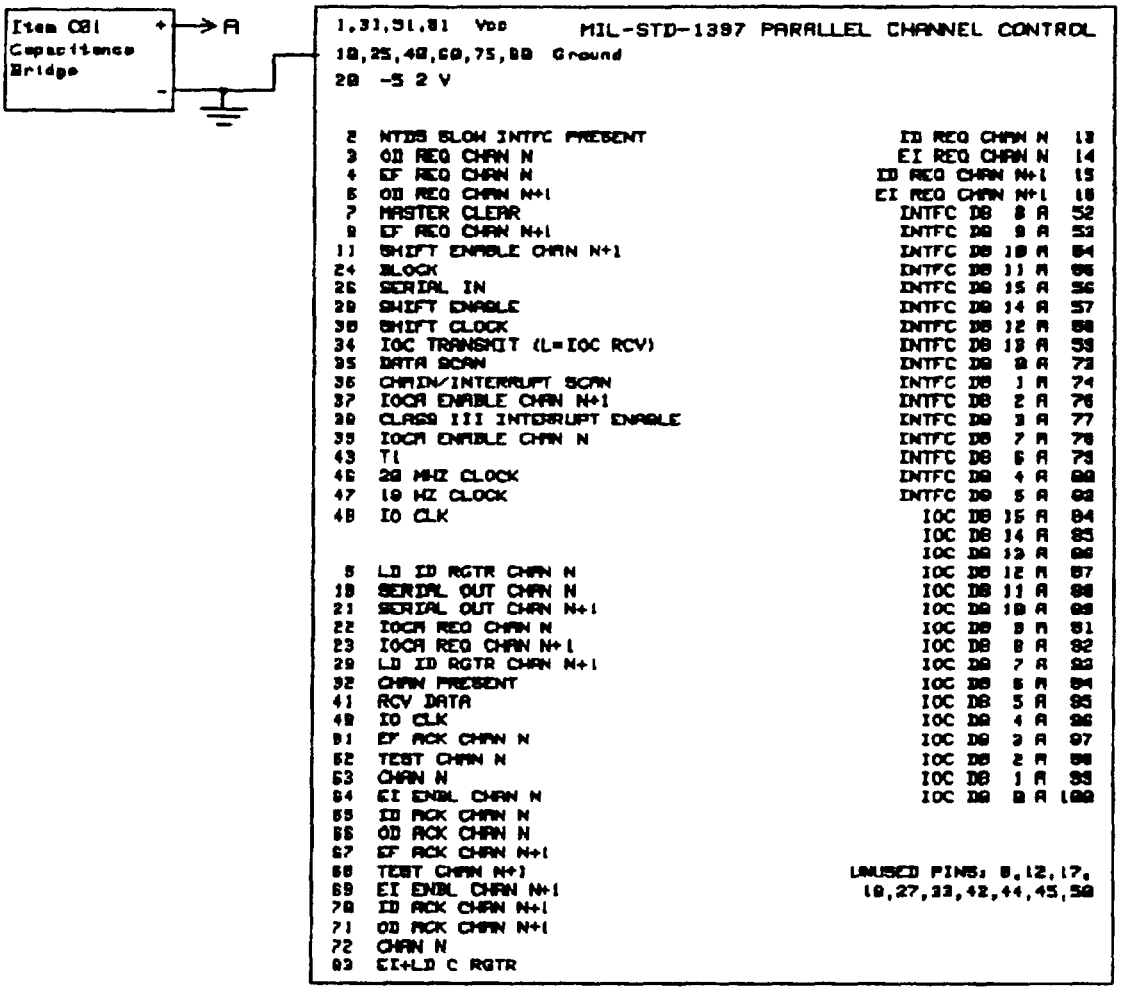


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31 51,81	164	393	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.589 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 11.4 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (32,768 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (32,768 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.589 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

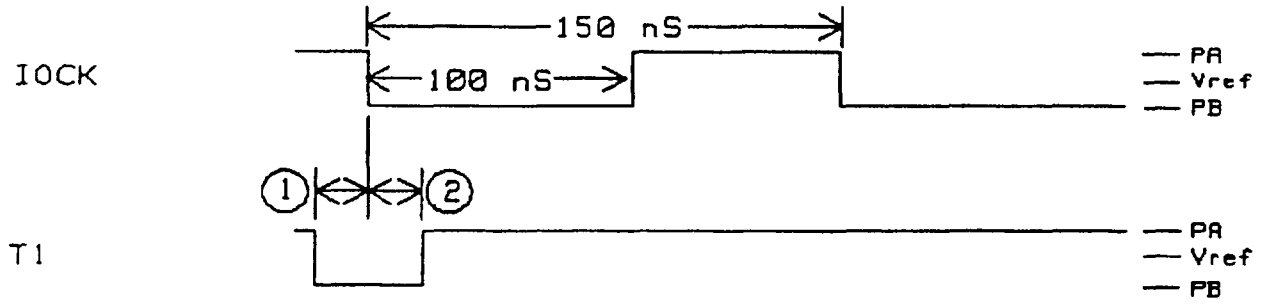
(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. (1) T1 to IOCK. Setup time: 25 nS minimum.
2. (2) IOCK to T1. Hold time: 4 nS minimum.

FIGURE 11. Input timing parameters for table XV.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (32,768 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (32,768 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (32,768 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (32,768 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.589 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

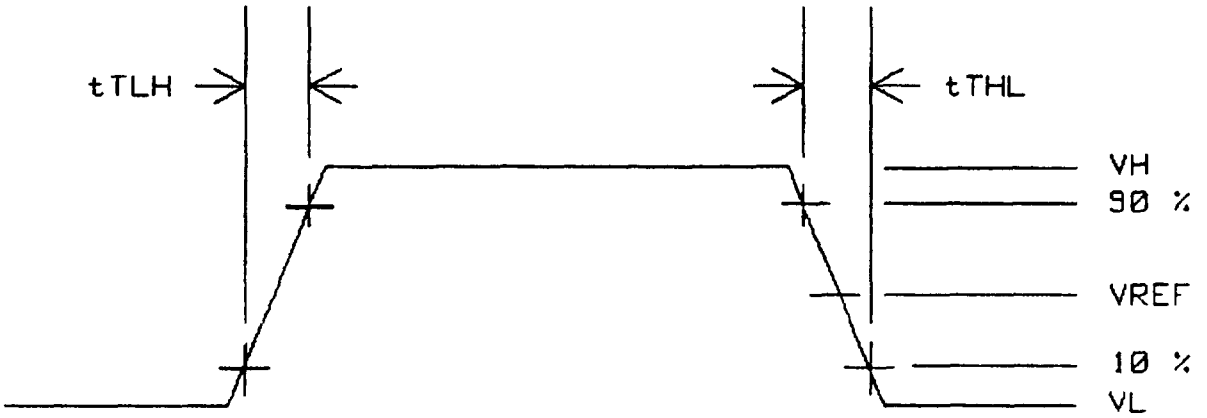
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. I/O waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1 1/	tpd	IOCK to IOC 0	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 1	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 2	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 3	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 4	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 5	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 6	-	-	-	86	86	86	nS	12
1	tpd	IOCK to IOC 7	-	-	-	86	86	86	nS	12
2 2/	tpd	IOCK to IOC 0	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 1	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 2	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 3	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 4	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 5	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 6	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 7	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 8	-	-	-	70	70	70	nS	12

1/ These signals contain control information.
2/ These signals contain data.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
2	tpd	IOCK to IOC 9	-	-	-	70	70	70	nS	12
2/ 2	tpd	IOCK to IOC 10	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 11	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 12	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 13	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 14	-	-	-	70	70	70	nS	12
2	tpd	IOCK to IOC 15	-	-	-	70	70	70	nS	12
3	tpd	IOCK to INTFC 0	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 1	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 2	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 3	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 4	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 5	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 6	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 7	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 8	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 9	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 10	-	-	-	73	73	73	nS	12

2/ These signals contain data.

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
3	tpd	IOCK to INTFC 11	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 12	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 13	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 14	-	-	-	73	73	73	nS	12
3	tpd	IOCK to INTFC 15	-	-	-	73	73	73	nS	12
4	tpd	IOCK to IOCAREQCHN	-	-	-	83	83	83	nS	12
4	tpd	IOCK to IOCAREQCHN+1	-	-	-	83	83	83	nS	12
5	tpd	IOCK to IDACKCHN	-	-	-	110	110	110	nS	12
5	tpd	IOCK to IDACKCHN+1	-	-	-	110	110	110	nS	12
6	tsu	T1 to IOCK	25	25	25	-	-	-	nS	11,12
7	th	IOCK to T1	4	4	4	-	-	-	nS	11,12
8	tpd	20MCK to ODACKCHN	-	-	-	103	103	103	nS	12
8	tpd	20MCK to ODACKCHN+1	-	-	-	103	103	103	nS	12
8	tpd	20MCK to EFACKCHN	-	-	-	103	103	103	nS	12
8	tpd	20MCK to EFACKCHN+1	-	-	-	103	103	103	nS	12

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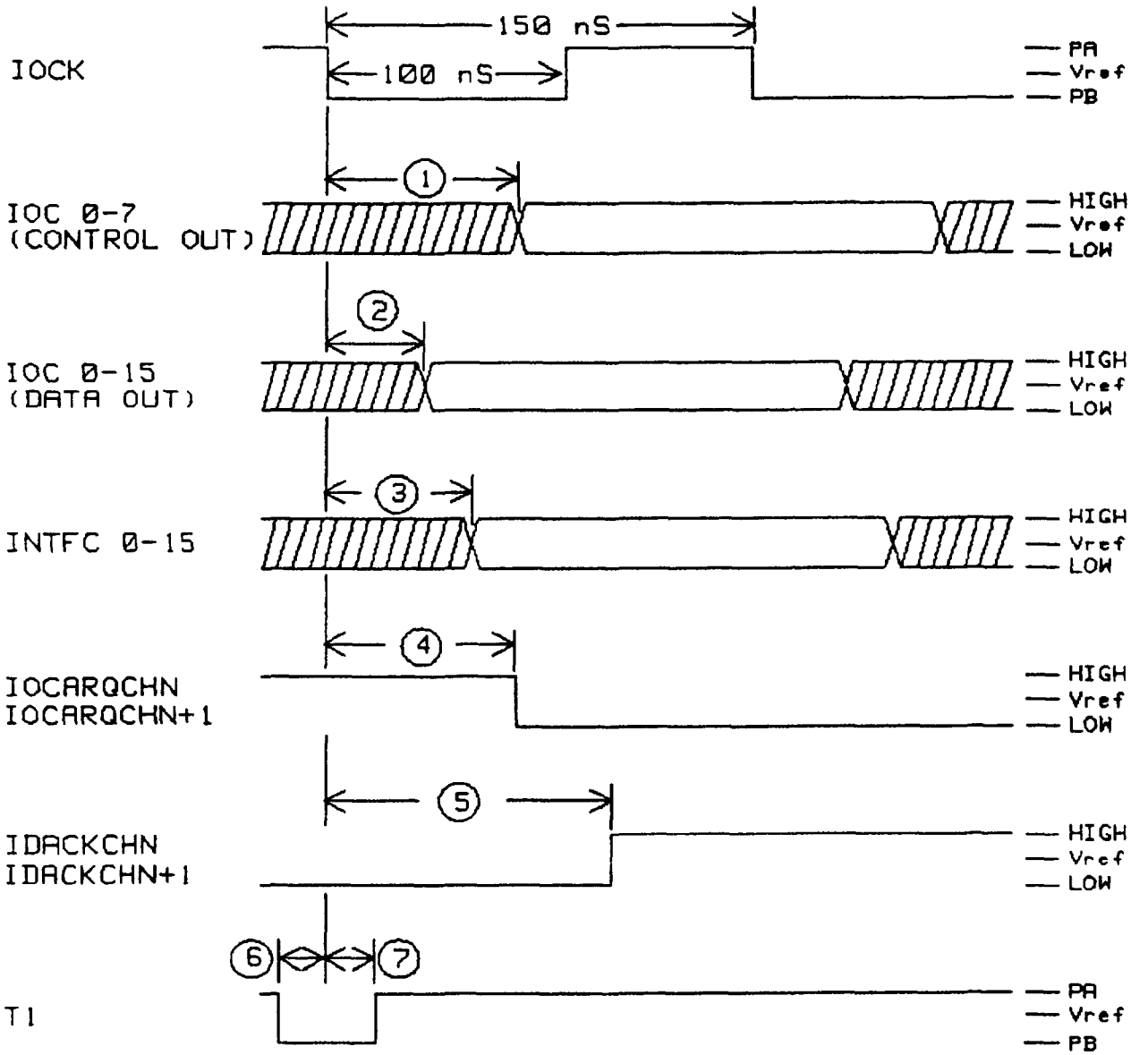


FIGURE 12a. Dynamic waveforms for table XVII.

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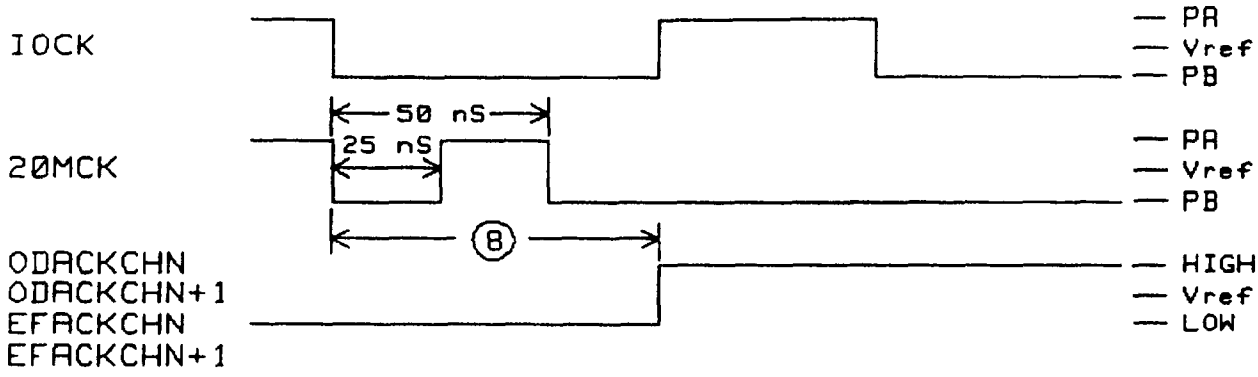


FIGURE 12b. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	NTDSSLOW	2	4	100	pF	1-4
R2	ODREQCHN	3	2	75	pF	1-4
R3	EFREQCHN	4				
R4	ODREQCHN+1	6				
R5	MACLR	7				
R6	EFREQCHN+1	9	2	75	pF	1-4
R7	BLOCK	24	4	75	pF	1-4
R8	SERIN	26				
R9	SHIFTEN	28				
R10	SHIFTCK	30	4	75	pF	1-4
R11	IOCXMIT	34	2	75	pF	1-4
R12	DTSCN	35				
R13	CHN/INTSCN	36	2	75	pF	1-4
R14	IOCAENCHN+1	37	4	100	pF	1-4
R15	CLIII INTEN	38	2	75	pF	1-4
R16	IOCAENCHN	39	4	100	pF	1-4
R17	T1	43	2	75	pF	1-4
R18	20MCK	46				
R19	19CK	47	2	75	pF	1-4
R20	IOCK	48	4	100	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
Bidirectionals						
R21	IOC 15	84	4	100	pF	1-4
R22	IOC 14	85	4	100	pF	1-4
R23	IOC 13	86	4	75	pF	1-4
R24	IOC 12	87	4	100	pF	1-4
R25	IOC 11	88	4	75	pF	1-4
R26	IOC 10	89	4	75	pF	1-4
R27	IOC 9	91	4	100	pF	1-4
R28	IOC 8	92	4	100	pF	1-4
R29	IOC 7	93	4	75	pF	1-4
R30	IOC 6	94				
R31	IOC 5	95				
R32	IOC 4	96				
R33	IOC 3	97				
R34	IOC 2	98				
R35	IOC 1	99				
R36	IOC 0	100	4	75	pF	1-4
R37	INTFC 8	52	4	100	pF	1-4
R38	INTFC 9	53				
R39	INTFC 10	54	4	100	pF	1-4
R40	INTFC 11	55	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
R41	INTFC 15	56	4	75	pF	1-4
R42	INTFC 14	57				
R43	INTFC 12	58				
R44	INTFC 13	59	4	75	pF	1-4
R45	INTFC 0	73	4	125	pF	1-4
R46	INTFC 1	74	4	100	pF	1-4
R47	INTFC 2	76				
R48	INTFC 3	77				
R49	INTFC 7	78				
R50	INTFC 6	79				
R51	INTFC 4	80				
R52	INTFC 5	82	4	100	pF	1-4
Input						
R53	SHIFTENCHN+1	11	4	75	pF	1-4
Bidirectionals						
R54	IDREQCHN	13	6	75	pF	1-4
R55	EIREQCHN	14	4	75	pF	1-4
R56	IDREQCHN+1	15	8	100	pF	1-4
R57	EIREQCHN+1	16	6	100	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R57
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R57
Capacitance measurement	NA	± 4.8 pF	R1
	NA	± 3.65 pF	R2-R6
	NA	± 3.55 pF	R7-R10
	NA	± 3.65 pF	R11-R13
	NA	± 4.8 pF	R14
	NA	± 3.65 pF	R15
	NA	± 4.8 pF	R16
	NA	± 3.65 pF	R17-R19
	NA	± 4.8 pF	R20-R22
	NA	± 3.55 pF	R23
	NA	± 4.8 pF	R24
	NA	± 3.55 pF	R25,R26
	NA	± 4.8 pF	R27,R28
	NA	± 3.55 pF	R29-R36
	NA	± 4.8 pF	R37-R39
	NA	± 3.55 pF	R40-R44
	NA	± 6.05 pF	R45
	NA	± 4.8 pF	R46-R52
	NA	± 3.55 pF	R53
	NA	± 3.45 pF	R54
	NA	± 3.55 pF	R55
	NA	± 4.6 pF	R56
	NA	± 4.7 pF	R57

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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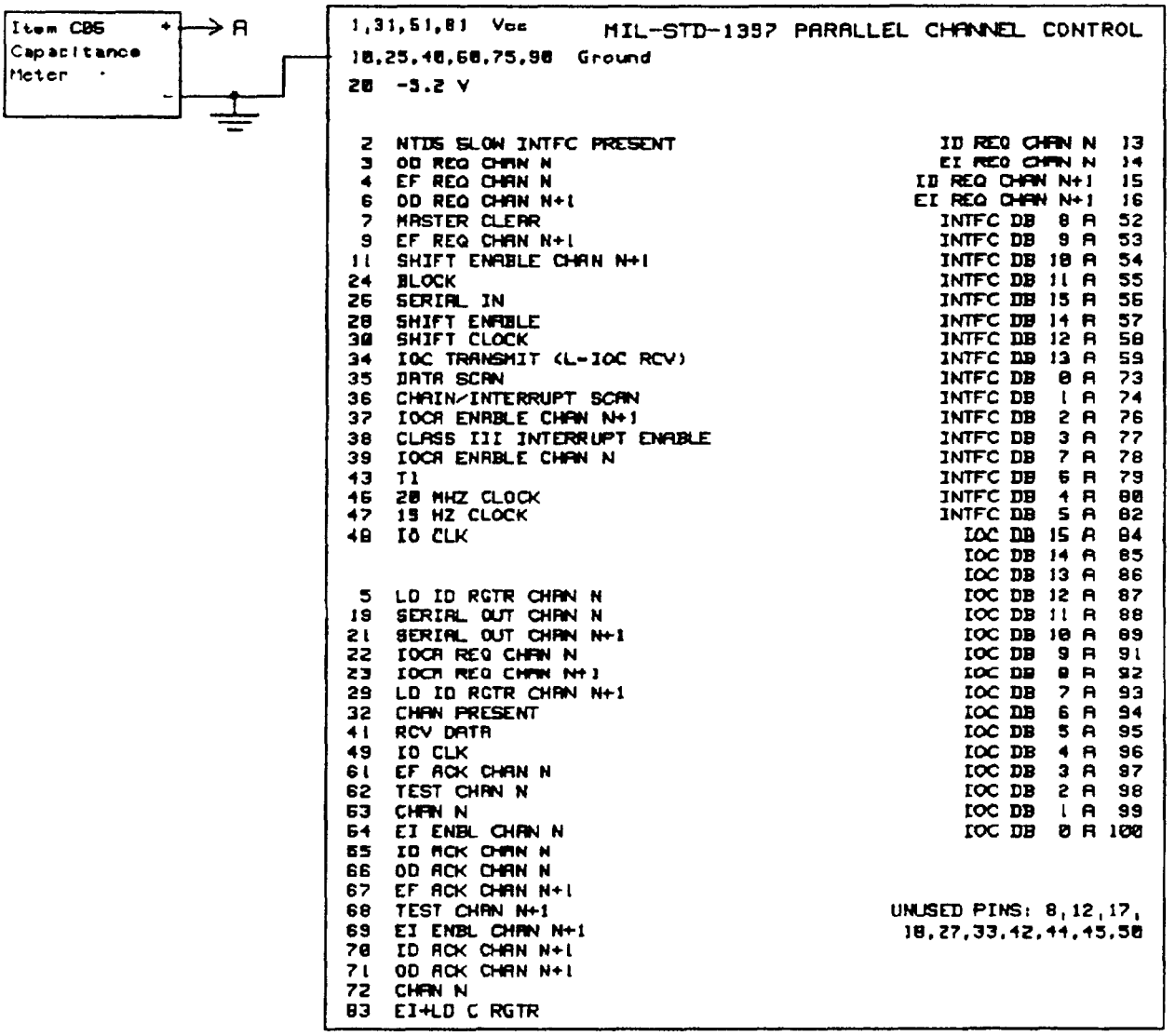


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
T	PSTA	1	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T
Power supply current limit	0.589 A	NA	T
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T T
Power supply transient amplitude measurement	NA	± 20 mV	T

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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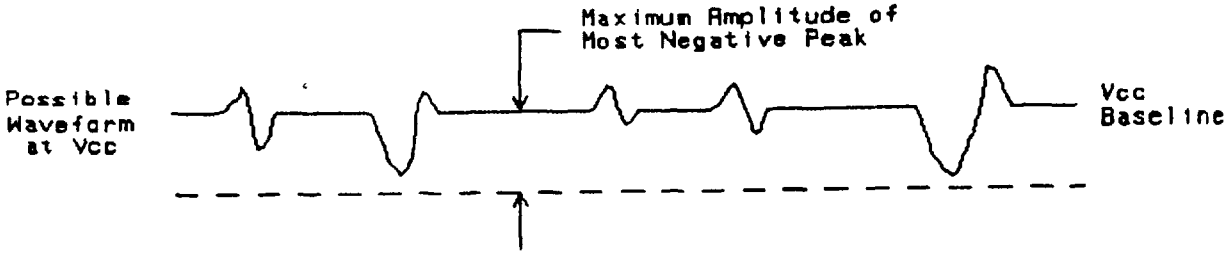
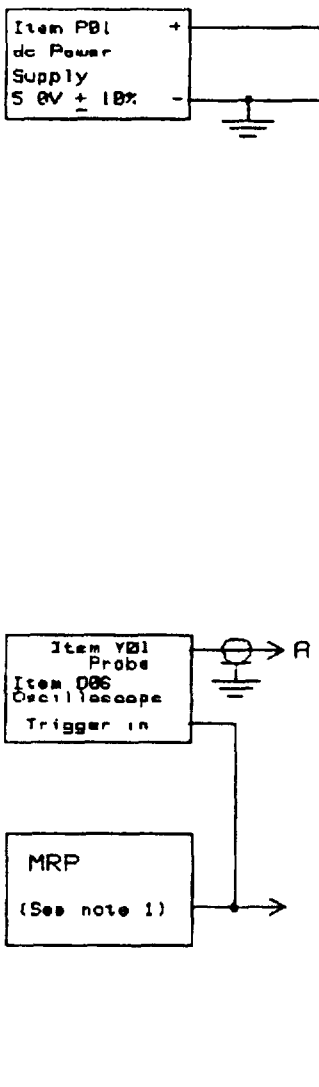


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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1,31,51,81	Vcc	MIL-STD-1397 PARALLEL CHANNEL CONTROL	
10,25,40,60,75,90	Ground		
20	-5.2 V		
2	NTDS SLOW INTFC PRESENT	ID REQ CHAN N	13
3	OD REQ CHAN N	EI REQ CHAN N	14
4	EF REQ CHAN N	ID REQ CHAN N+1	15
6	OD REQ CHAN N+1	EI REQ CHAN N+1	16
7	MASTER CLEAR	INTFC DB 8 A	52
8	EF REQ CHAN N+1	INTFC DB 9 A	53
11	SHIFT ENABLE CHAN N+1	INTFC DB 10 A	54
24	BLOCK	INTFC DB 11 A	55
26	SERIAL IN	INTFC DB 15 A	56
28	SHIFT ENABLE	INTFC DB 14 A	57
30	SHIFT CLOCK	INTFC DB 12 A	58
34	IOC TRANSMIT (L-IOC RCV)	INTFC DB 13 A	59
35	DATA SCAN	INTFC DB 0 A	73
36	CHAIN/INTERRUPT SCAN	INTFC DB 1 A	74
37	IOCA ENABLE CHAN N+1	INTFC DB 2 A	75
38	CLASS III INTERRUPT ENABLE	INTFC DB 3 A	77
39	IOCA ENABLE CHAN N	INTFC DB 7 A	78
43	T1	INTFC DB 6 A	79
46	20 MHZ CLOCK	INTFC DB 4 A	80
47	10 HZ CLOCK	INTFC DB 5 A	82
48	IO CLK	IOC DB 15 A	84
		IOC DB 14 A	85
		IOC DB 13 A	86
5	LD ID RGTR CHAN N	IOC DB 12 A	87
19	SERIAL OUT CHAN N	IOC DB 11 A	88
21	SERIAL OUT CHAN N+1	IOC DB 10 A	89
22	IOCA REQ CHAN N	IOC DB 9 A	91
23	IOCA REQ CHAN N+1	IOC DB 8 A	92
29	LD ID RGTR CHAN N+1	IOC DB 7 A	93
32	CHAN PRESENT	IOC DB 6 A	94
41	RCV DATA	IOC DB 5 A	95
43	IO CLK	IOC DB 4 A	96
61	EF ACK CHAN N	IOC DB 3 A	97
62	TEST CHAN N	IOC DB 2 A	98
63	CHAN N	IOC DB 1 A	99
64	EI ENBL CHAN N	IOC DB 0 A	100
65	ID ACK CHAN N		
66	OD ACK CHAN N		
67	EF ACK CHAN N+1		
68	TEST CHAN N+1	UNUSED PINS: 9,12,17,	
69	EI ENBL CHAN N+1	18,27,33,42,44,45,50	
70	ID ACK CHAN N+1		
71	OD ACK CHAN N+1		
72	CHAN N		
83	EI+LD C RGTR		

NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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1. DOCUMENT NUMBER	2 DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

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a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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